

The Dynamic Complementarity Tax: Two Consumers, One Record, and an Aging Reference

Paper VII skeleton — v0.1, for venue work

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Abstract

When two consumers read one stored record, the joint description costs more than the better single-consumer description — a complementarity tax, on both the rate and the ideal reset-work faces. We characterize how that tax responds when the eraser’s reference ages: every observation access class enters through one conditional variance (equal- q classes pay identical taxes — a universality that we prove is *exactly* a Gaussian privilege and only approximately a binary one); the tax’s staleness response obeys an envelope sign law (it rises iff the joint record couples to the context harder than the binding consumer, with an always-downward kink at the consumer switch); the whole theory lifts to jointly stationary processes as per-frequency matrix water-level systems under exactly two common Lagrange prices; and a new m -record moment-convexity lemma makes the underlying weighted program convex, resolving uniqueness and making the two-price allocation globally optimal. A decode-threshold experiment measures the rising tax at +0.090 against a +0.086 prediction, with an equal- q analytic-equality control at 0.001. [TODO: compress per venue.]

1 Introduction and related work

[TODO: Attribution spine from go13-dynamic-tax-NOVELTY.md: Xiao–Luo 2005 (static kernel), Stylianou–Charalambous 2021/2024 (tuple reduction, implicit static two-water-level system — CRITICAL: also scopes the static substrate’s rate face), HB/Kaspi + Timo–Chan–Grant, vending-machine line. RE-SWEEP the Stylianou–Charalambous line immediately pre-submission.]

2 The static substrate

[TODO: Import GO-11 Thms 5/9 (statements only, cite the GO-11 manuscript); state the 074 uniqueness resolution here or in §7.]

3 The matrix- q reduction and access-class universality

[TODO: Theorem 1 + the $r \geq 2$ counterexample scoping.]

4 The tax-curve sign law

[TODO: Theorem 2 (envelope; downward kink; $w=1$ flat; $CT \geq 0$); the exploratory regime map as a disclosed figure; the 067 reference-disambiguation story as a caution box.]

5 The binary twin

[TODO: Theorem 3: slice collapse; reliability-distribution closed form; generalized tilt; EXACT non-collapse at 50-digit precision vs near-universality (envelope $\leq 2.1 \times 10^{-4}$, observation-grade) — the paper’s most quotable dichotomy.]

6 The spectral $m = 2$ theorem

[TODO: Theorem 4: per-frequency Theorem-9 systems under two common prices; the “pair of prices” vindication narrative (refuted at $m = 1$, exact at $m = 2$); circulant scope + Toeplitz debts shared with Paper VI.]

7 The moment-convexity lemma

[TODO: Theorem 5 in full (four-step proof; $G = -\log \det(I - Z)$; Hessian decomposition; flats; uniqueness scoping with the shut-off-record failure mode). CANDIDATE: split out as a standalone short communication if the venue prefers a leaner Paper VII.]

8 The operational face

[TODO: 069: instrument lineage, design frozen pre-instrument, measured $+0.090$ vs $+0.086$, equal- q control 0.001 ; the V6 instrumentation amendment disclosed (both governed runs’ physics 5/5).]

Reproducibility

Sealed preregistrations 067–069, 072–074; governed harnesses CI-re-run on every push; fresh-context adversarial verification for every theorem (two draft refutations caught pre-seal, on record). Map: PAPER-VI-VII-EVIDENCE-INDEX.md.

[TODO: Bibliography per the attribution spine.]